

Machine Learning Models Evaluation Results

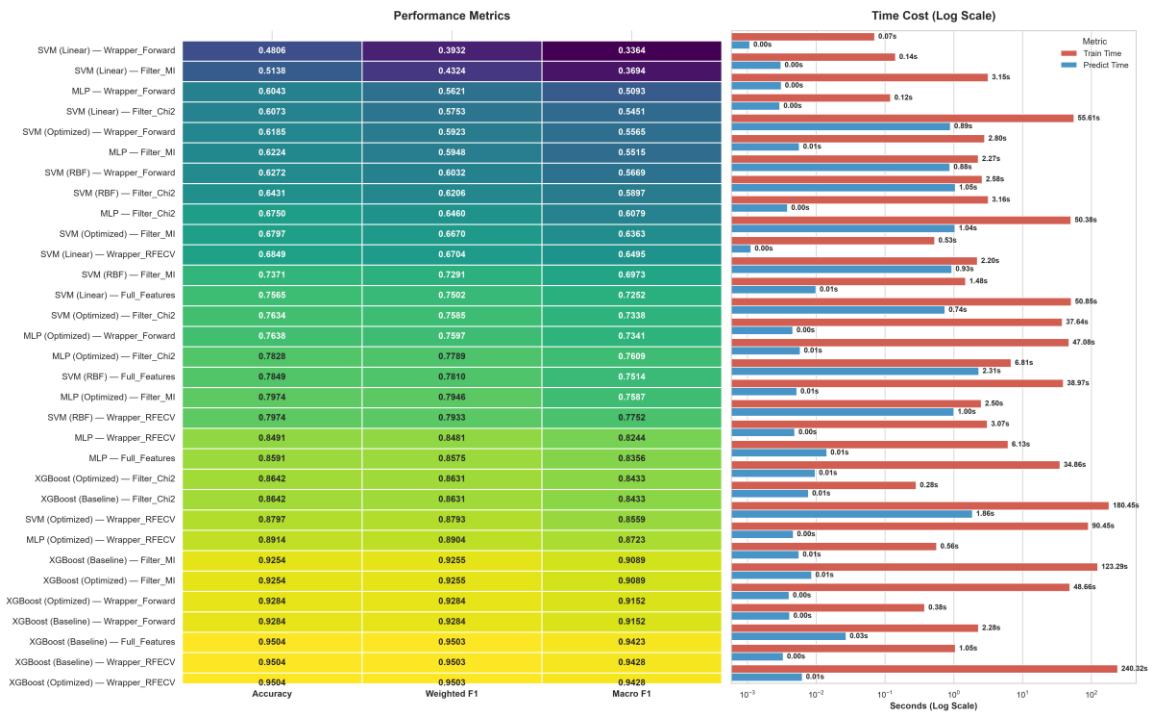
Final Results Summary (Sorted by Accuracy)

Algorithm	Accuracy	Weighted F1	Macro F1	Train Time	Predict Time
XGBoost (Baseline) — Full_Features	0.9504	0.9503	0.9423	2.28s	0.03s
XGBoost (Baseline) — Wrapper_RFECV	0.9504	0.9503	0.9428	1.05s	0.00s
XGBoost (Optimized) — Wrapper_RFECV	0.9504	0.9503	0.9428	240.32s	0.01s
XGBoost (Optimized) — Wrapper_Forward	0.9284	0.9284	0.9152	48.66s	0.00s
XGBoost (Baseline) — Wrapper_Forward	0.9284	0.9284	0.9152	0.38s	0.00s
XGBoost (Baseline) — Filter_MI	0.9254	0.9255	0.9089	0.56s	0.01s
XGBoost (Optimized) — Filter_MI	0.9254	0.9255	0.9089	123.29s	0.01s
MLP (Optimized) — Wrapper_RFECV	0.8914	0.8904	0.8723	90.45s	0.00s
SVM (Optimized) — Wrapper_RFECV	0.8797	0.8793	0.8559	180.45s	1.86s
XGBoost (Optimized) — Filter_Chi2	0.8642	0.8631	0.8433	34.86s	0.01s
XGBoost (Baseline) — Filter_Chi2	0.8642	0.8631	0.8433	0.28s	0.01s
MLP — Full_Features	0.8591	0.8575	0.8356	6.13s	0.01s
MLP — Wrapper_RFECV	0.8491	0.8481	0.8244	3.07s	0.00s
MLP (Optimized) — Filter_MI	0.7974	0.7946	0.7587	38.97s	0.01s

SVM (RBF) — Wrapper_RFECV	0.7974	0.7933	0.7752	2.50s	1.00s
SVM (RBF) — Full_Features	0.7849	0.7810	0.7514	6.81s	2.31s
MLP (Optimized) — Filter_Chi2	0.7828	0.7789	0.7609	47.08s	0.01s
MLP (Optimized) — Wrapper_Forward	0.7638	0.7597	0.7341	37.64s	0.00s
SVM (Optimized) — Filter_Chi2	0.7634	0.7585	0.7338	50.85s	0.74s
SVM (Linear) — Full_Features	0.7565	0.7502	0.7252	1.48s	0.01s
SVM (RBF) — Filter_MI	0.7371	0.7291	0.6973	2.20s	0.93s
SVM (Linear) — Wrapper_RFECV	0.6849	0.6704	0.6495	0.53s	0.00s
SVM (Optimized) — Filter_MI	0.6797	0.6670	0.6363	50.38s	1.04s
MLP — Filter_Chi2	0.6750	0.6460	0.6079	3.16s	0.00s
SVM (RBF) — Filter_Chi2	0.6431	0.6206	0.5897	2.58s	1.05s
SVM (RBF) — Wrapper_Forward	0.6272	0.6032	0.5669	2.27s	0.88s
MLP — Filter_MI	0.6224	0.5948	0.5515	2.80s	0.01s
SVM (Optimized) — Wrapper_Forward	0.6185	0.5923	0.5565	55.61s	0.89s
SVM (Linear) — Filter_Chi2	0.6073	0.5753	0.5451	0.12s	0.00s
MLP — Wrapper_Forward	0.6043	0.5621	0.5093	3.15s	0.00s
SVM (Linear) — Filter_MI	0.5138	0.4324	0.3694	0.14s	0.00s
SVM (Linear) — Wrapper_Forward	0.4806	0.3932	0.3364	0.07s	0.00s

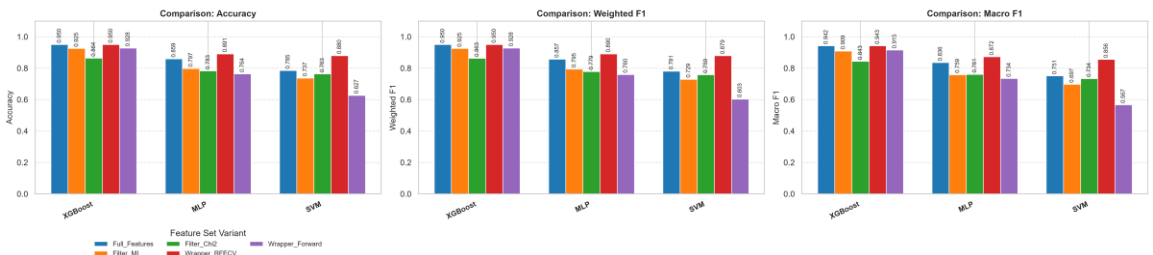
Comprehensive Model Overview

Comprehensive Model Overview: Performance & Computational Cost



Algorithm vs. Feature Selection Comparison

Algorithm vs. Feature Selection Comparison

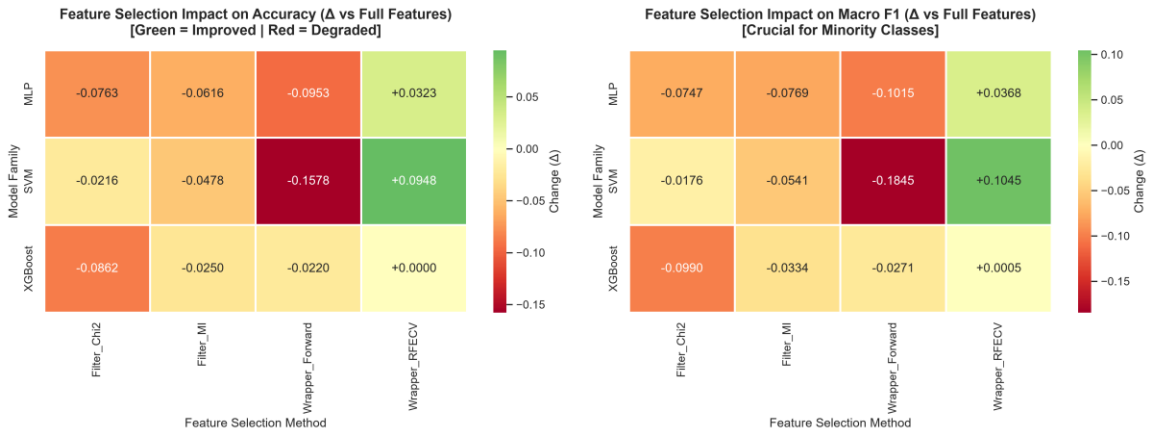


Filter Impact vs. Full-Features Baseline

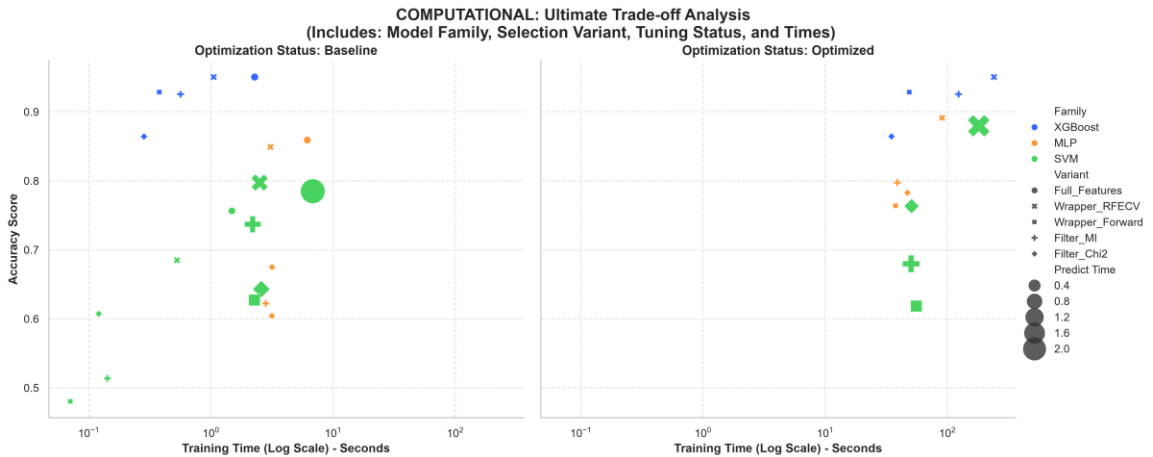
Model Family	Feature Set	Δ Accuracy	Δ Macro F1	Train Time	Predict Time
XGBoost	Filter_MI	-0.0250	-0.0334	0.56s	0.0056s
XGBoost	Filter_Chi2	-0.0862	-0.0990	34.86s	0.0096s
XGBoost	Wrapper_RFECV	+0.0000	+0.0005	1.05s	0.0033s
XGBoost	Wrapper_Forward	-0.0220	-0.0271	48.66s	0.0040s

MLP	Filter_MI	-0.0616	-0.0769	38.97s	0.0052s
MLP	Filter_Chi2	-0.0763	-0.0747	47.08s	0.0058s
MLP	Wrapper_RFECV	+0.0323	+0.0368	90.45s	0.0046s
MLP	Wrapper_Forward	-0.0953	-0.1015	37.64s	0.0046s
SVM	Filter_MI	-0.0478	-0.0541	2.20s	0.9343s
SVM	Filter_Chi2	-0.0216	-0.0176	50.85s	0.7427s
SVM	Wrapper_RFECV	+0.0948	+0.1045	180.45s	1.8621s
SVM	Wrapper_Forward	-0.1578	-0.1845	2.27s	0.8774s

Feature Selection Impact Heatmaps

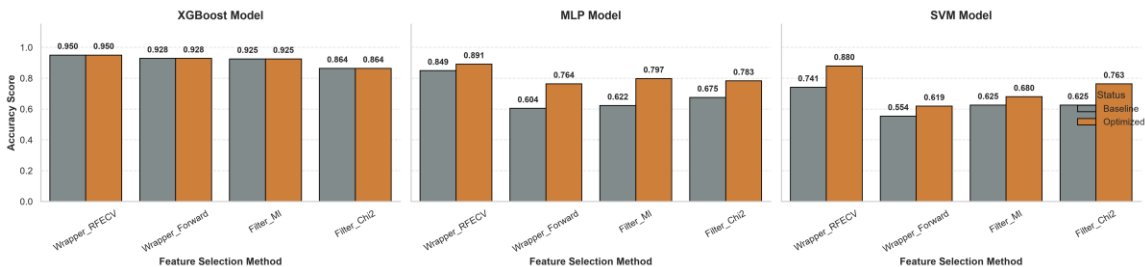


Computational vs Accuracy Trade-off Analysis



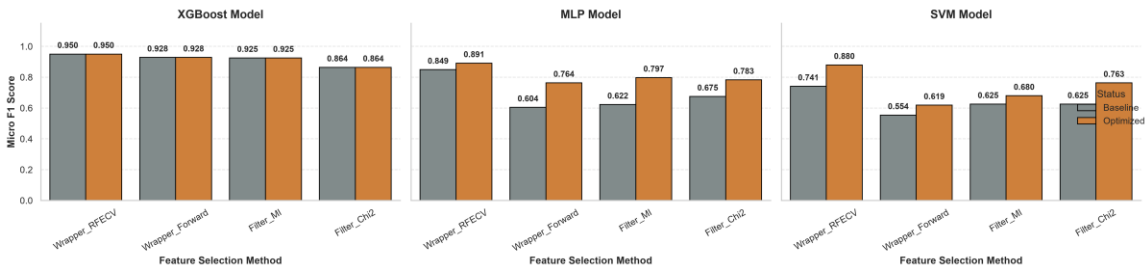
GridSearchCV Impact on Accuracy

Impact of GridSearchCV on [Accuracy] across Feature Selection Methods



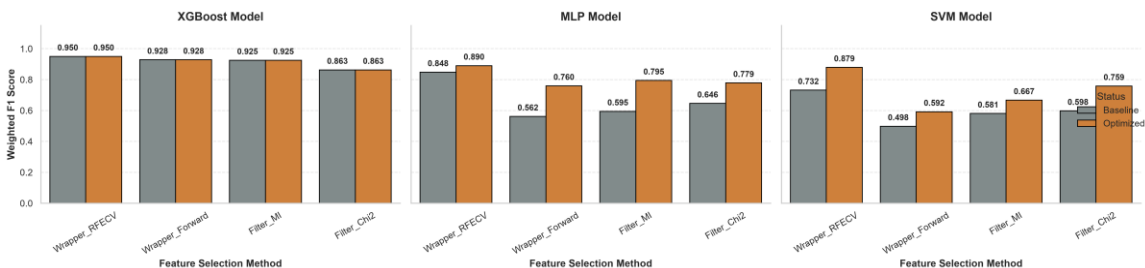
GridSearchCV Impact on Micro F1

Impact of GridSearchCV on [Micro F1] across Feature Selection Methods



GridSearchCV Impact on Weighted F1

Impact of GridSearchCV on [Weighted F1] across Feature Selection Methods



Overall Impact of Hyperparameter Tuning

